TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Freestanding, Ultrathin Micro-LED Technology for Holographic Waveguide-Compatible Microdisplay Light Engine			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( ) *(Please choose at least one of the areas above)*			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical・Electronics		Electrical・Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Display	Display Components and Materials	Semiconductor Devices and Systems	Compound Semiconductor Devices
Concept of Proposed Technology	This proposal introduces **ultrathin nitride μLED membranes** grown on 2D/sapphire substrate. The atomically thin 2D interlayer, such as graphene or h-BN layer, facilitates damage-free lift-off, enabling the transfer of high-quality epilayers without the physical degradation associated with conventional laser lift-off. A critical feature is the precise control of the membrane thickness to approximately 600 nm, which forms a vertical Fabry-Perot cavity. This structure significantly narrows the spectral linewidth, thereby maximizing coupling efficiency into Bragg-selective waveguides. The technology further encompasses **monolithic 3D integration**, where the released membrane can be directly bonded to a **Si CMOS backplane** and subsequently pixelated via standard semiconductor lithography. This process eliminates the alignment errors inherent in conventional chip-transfer methods, enabling ultra-high-density pixel arrays. Finally, a monolithically integrated optical elements collimate the emission to match the narrow angular acceptance cone of the waveguide, realizing compact, high-fidelity holographic augmented-reality (AR) display systems.			

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Rationale for Joint R&D Project	The UIUC research team possesses core technology for manufacturing detachable ultrathin LED epi-membranes, a core material for next-generation AR displays. Transforming this material innovation into a fully functional holographic display system requires integration capabilities that extend beyond material synthesis. Specifically, the realization of a driving module involves precise heterogeneous bonding to CMOS backplanes, sub-micron lithographic pixel definition, and advanced electrical optical co-packaging. These processes demand industrial-grade semiconductor infrastructure and manufacturing know-how. Therefore, collaboration with Korean SMEs possessing established CMOS integration and system realization capabilities is indispensable. By synergizing UIUC's source material expertise with the partner's manufacturing process technology, this joint project aims to bridge the gap between laboratory research and commercialization, successfully developing a manufacturing-ready, ultra-compact holographic light engine.
Keywords	1. micro-LED 2. remote epitaxy 3. quasi-van der Waals epitaxy 4. Si CMOS backplane circuitry 5. holographic display 6. Advanced 3D Packaging

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2)]	**Start of the Project**	Level (3)	**End of the Project**	Level (5/6)
R&D Objectives	**1. Introduction of μLED technologies** μLED technologies, which employ inorganic III-N and/or III-V diodes as light emitters, are widely considered a game changer for next-generation displays, due to their inherent performance advantages over organic LEDs (OLEDs) and liquid crystal displays (LCDs) in achievable brightness, efficiency, contrast ratio, and lifetime. [1] For example, μLED-based displays can reach peak luminance levels exceeding several million nits, enabling outdoor readability and augmented/virtual reality (AR/VR) applications, while their inorganic nature ensures superior environmental and thermal stability. However, despite these compelling advantages, the realization of μLED-based systems faces several critical challenges, particularly in achieving ultrahigh pixel densities while preserving device efficiency and scalability. One fundamental bottleneck arises from materials and device scaling limitations at the microscale. As pixel dimensions shrink below a few micrometers, maintaining high external quantum efficiency across red, green, and blue (RGB) emitters becomes increasingly difficult. Beyond material-related constraints, system-level integration presents another major obstacle. State-of-the-art nitride μLEDs are typically grown on sapphire substrates, necessitating lift-off and subsequent pick-and-place transfer processes for integration with thin-film transistor (TFT) backplanes. While numerous transfer strategies have been explored, current approaches face inherent limitations in scalability, yield, throughput, and placement accuracy, particularly when millions of sub-micron RGB pixels must be assembled over large areas. These challenges severely hinder the realization of ultrahigh-resolution displays and impose substantial manufacturing complexity and cost. In addition to these integration challenges, the prevalent top-top contact device architecture, dictated by the inaccessibility of the epilayer backside prior to substrate removal, introduces further performance limitations. In such configurations, both p- and n-type contacts are formed on the top surface, leading to non-uniform current injection and poor lateral current spreading, effects that become increasingly severe as device dimensions shrink. Moreover, the elongated current paths and			

parasitic resistances associated with top-top contact layouts result in increased RC delay, thereby limiting the modulation speed of μLEDs. This constraint is particularly detrimental for emerging applications that require high-speed operation, such as optical interconnects and active-matrix-driven microdisplays. Furthermore, the lack of backside access precludes the implementation of vertical device geometries and optical resonant cavities, which are widely employed in laser diodes and vertical LEDs to enhance light extraction efficiency, spectral purity, and modulation performance. As a result, conventional μLED architectures face intrinsic trade-offs between efficiency, speed, and scalability, underscoring the need for fundamentally new integration and device design paradigms.

2. Key technologies in PI's group

Our key technology is based on 2D materials-assisted epitaxy, including remote epitaxy and quasi-vdW epitaxy, which enables the growth of high-quality III-N and III-V membranes and their controlled exfoliation at wafer scales. This capability provides a scalable pathway for heterogeneous integration beyond the limitations of conventional substrate-bound μLED architectures.

2.1. 2D materials-assisted epitaxy technique for high-quality III-N films

The PI's studies revealed that when epitaxial growth on sp^2-bonded 2D surfaces, the interfacial energetics are fundamentally modified [2,3]. Specifically, the energy barrier for interfacial sliding is significantly lower than that required for dislocation nucleation, allowing misfit strain to be relieved through interfacial displacement rather than dislocation generation (Fig. 1).

Experimental evidence supporting this mechanism has been demonstrated across multiple compound semiconductor systems. For example, GaP grown on GaAs with a 4% lattice mismatch exhibits dramatically reduced dislocation densities under remote epitaxy conditions (Fig. 1). Similarly, GaN films grown on graphene/SiC substrates show substantially improved crystal quality, with threading dislocation densities reduced to ~2×10^7 cm^{-2}—significantly lower than those of GaN films grown directly on SiC substrates [4].

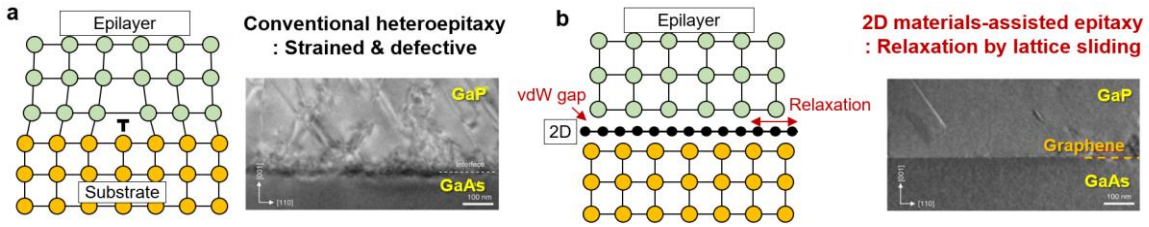


Figure 1. Impact of 2D materials-assisted epitaxy. a, Conventional heteroepitaxy: Epilayers are strained and defective (left: schematic, right: TEM). **b,** 2D materials enables strain relaxation without dislocation formation, dramatically improving the crystallinity and providing a path for long-wavelength, efficient III-N or III-V MQW.

This ability to form strain-relaxed, low-dislocation-density epilayers directly addresses a central materials bottleneck in the development of high-efficiency III-N and III-V μLEDs. In particular, emission wavelengths beyond 650 nm require indium compositions exceeding ~40%, a regime that has historically been inaccessible due to mismatch-driven degradation in crystalline quality. By enabling strain relaxation without dislocation formation, 2D materials-assisted epitaxy provides a viable pathway toward high-indium-content InGaN with preserved structural and optical integrity. Consistent with this expectation, preliminary studies from other groups have already demonstrated enhanced indium incorporation and emission extended into the red spectral range, validating the promise of this approach.

Compared with alternative strategies—such as porous templates, V-groove engineering, or nitride nanorod architectures—this thin-film-compatible, planar growth technique is uniquely suited for scalable device fabrication. Importantly, the resulting high-quality epilayers remain compatible with subsequent exfoliation and transfer processes, establishing a direct link between materials innovation and heterogeneous integration for ultrahigh density μLED arrays.

2.2. 2D materials-assisted epitaxy for layer transfer

Another critical advantage of 2D materials-assisted epitaxy is the capability to separate epitaxial

films cleanly at the 2D interface via simple mechanical exfoliation. Because the epitaxial layers are bound to the underlying 2D materials through weak van der Waals interactions, they can be released with atomic-level precision at dramatically lower energies than those required for conventional lift-off processes, such as laser lift-off or chemical etching.

In the context of μLED fabrication, this capability enables a fundamentally different integration pathway. Entire LED epilayers or pre-patterned μLED pixel arrays can be exfoliated and directly transferred onto CMOS or TFT backplanes, effectively bypassing the yield, throughput, and placement-accuracy limitations inherent to conventional pick-and-place or mass-transfer techniques. This layer-level transfer approach is particularly well suited for ultrahigh-density μLED arrays, where the individual manipulation of millions of microscale pixels becomes impractical.

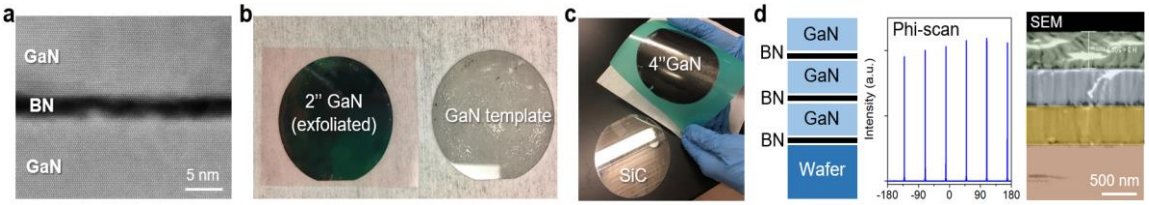
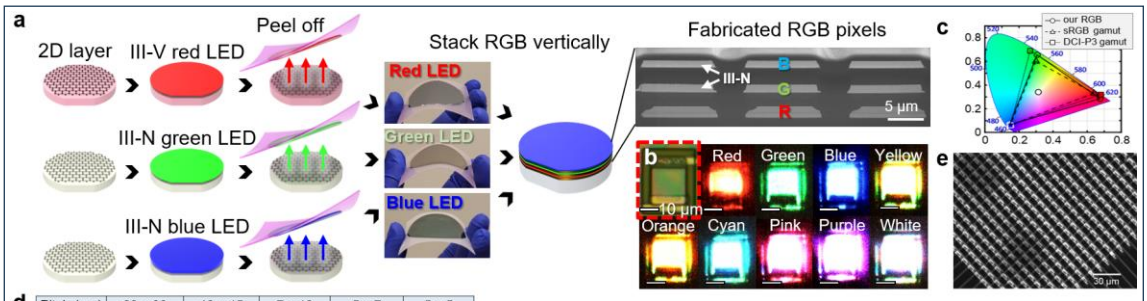


Figure 2. a, TEM of GaN remote epitaxy on 2D BN. **b,** wafer-scale exfoliation of GaN(0001) thin films from BN/GaN substrate and **c,** graphene/SiC substrate. **d,** *In situ* growth of three GaN/BN stacks. XRD and cross-sectional SEM confirming single-crystal GaN stacks with BN interlayers.

The PI has pioneered wafer-scale growth of 2D materials on substrates compatible with nitride epitaxy, including large-area boron nitride (BN) films grown on GaN substrates at relatively low temperatures (500–700 ℃), as well as graphene formed on SiC. As shown in Fig. 2, GaN epilayers grown on these 2D templates via remote epitaxy can be mechanically exfoliated precisely at the 2D interface over wafer-scale areas, directly validating the scalability and uniformity of the process.

Importantly, because both BN and III-N layers can be grown within the same MBE or MOCVD reactor environment, *in-situ* growth of alternating 3D/2D heterostructures become feasible without breaking vacuum. To demonstrate this capability, the PI achieved *in-situ* growth of GaN/BN multilayer stacks (GaN/BN/GaN/BN/GaN/BN) in a single epitaxy run, followed by sequential exfoliation of individual GaN layers (Fig. 2d). This result established a robust proof of concept for vertically stacked nitride LED structures incorporating BN interlayers, offering unprecedented control over layer thickness, stacking order, and release—capabilities that are central to the scalable heterogeneous integration strategy pursued in this project.

2.3. Preliminary demonstration: Vertically stacked 5,100 ppi full-color LEDs enabled by 2D materials-assisted epitaxy



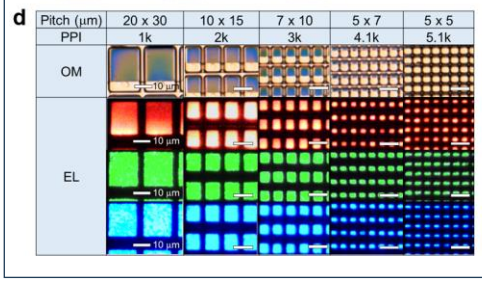


Figure 3. Demonstration of vertically stacked ultrahigh-ppi mLED. a, Fabrication of vertical RGB LEDs via RGB layer exfoliation from 2D and stacking. **b,** Optical microscope image of stacked RGB pixel operation and **c,** chromacity of vertical LEDs. **d,** Demonstration of full-color LEDs of up to 5,100 ppi enabled by our approach. **e,** SEM image of LEDs (single color) originally grown on 2D templates and then integrated with Si TFT backplane for active matrix operation.

The capabilities described above—namely, the formation of high-quality, strain-relaxed epitaxial membranes via 2D materials–assisted epitaxy and their controlled mechanical exfoliation—unlock entirely new nitride device architectures that were previously inaccessible. Leveraging these capabilities, the PI has demonstrated full-color μLED displays with record-small pixel sizes and ultrahigh pixel densities, establishing a compelling proof of concept for the proposed integration paradigm [5].

In these demonstrations, GaN/InGaN MQW blue and green LEDs were grown on BN–coated sapphire substrates, while III–As/P-based red LEDs were grown on graphene-coated GaAs substrates. Each epitaxial LED layer was subsequently exfoliated precisely at the 2D interface (Fig. 3a) and vertically stacked to form a monolithically aligned RGB architecture. To suppress optical crosstalk between stacked emitters, color filter layers were inserted between the RGB components. The vertically stacked structure was then patterned into individual RGB pixels through top-down etching and metallization processes (Fig. 3b,c).

In contrast to conventional pick-and-place approaches—which rely on the sequential transfer and alignment of individual red, green, and blue subpixels—this layer-level stacking strategy eliminates inter-subpixel alignment margins by assembling entire epitaxial layers prior to pixelation. As a result, pixel density can be dramatically increased without compromising the surface-to-volume ratio of the emitters. Moreover, the concept of vertically stacked RGB subpixels offers a theoretical density enhancement of up to threefold compared with laterally arranged architectures. Using this approach, the PI achieved a record pixel pitch of 5 μm, corresponding to an ultrahigh pixel density of 5,100 ppi (Fig. 3d).

Beyond display resolution, the PI further demonstrated the direct integration of exfoliated μLED layers with silicon-on-insulator (SOI) TFT backplanes (Fig. 3e), confirming the feasibility of seamless μLED–CMOS integration enabled by 2D materials–assisted transfer. This result directly addresses one of the most critical bottlenecks in μLED manufacturing: scalable and precise integration with electronic driving circuitry.

Collectively, these preliminary results validate not only the technical feasibility of the proposed research, but also its transformative potential. The demonstrated combination of wafer-scale remote epitaxy, unconventional strain relaxation, atomic-precision exfoliation, and vertically stacked RGB integration establishes a robust foundation for the proposed research thrusts. Together, these advances uniquely position the PI to realize efficient, vertically integrated, and manufacturable μLED-based light engines, enabling new frontiers in ultrahigh-resolution AR/VR displays and beyond.

3. Research objectives

The overarching objective of this research is to **mature 2D materials–assisted epitaxy into a robust and scalable platform for next-generation nitride μLED technologies**, addressing both materials-level and integration-level limitations identified above. Specifically, the project will pursue the following objectives:

3.1. Maturation of nitride μLED growth on 2D materials

The first objective is to advance the growth of III-N and III-V μLED structures on 2D material templates toward device-grade maturity. This includes systematic optimization of crystalline quality and internal quantum efficiency (IQE), with a particular focus on suppressing defect formation and non-radiative recombination under aggressive device scaling. In parallel, precise thickness control will be established to realize ultrathin epitaxial stacks, including buffer layers and active regions with a total thickness at 500-1000 nm range. Such thickness control is essential for enabling membrane-based device architectures, efficient heat dissipation, and compatibility with heterogeneous integration.

3.2. Development of scalable exfoliation and integration processes

The second objective is to develop reliable and scalable exfoliation and bonding processes for transferring nitride μLED membranes onto electronic backplanes. Building on 2D-assisted exfoliation, this effort will focus on achieving high-yield, wafer-scale release and integration of μLED epilayers while preserving electrical and optical performance. Particular emphasis will be placed on bonding strategies compatible with silicon CMOS and TFT circuitry, enabling precise alignment, mechanical

	robustness, and low-resistance electrical interfacing. Together, these objectives aim to establish a complete materials-to-integration workflow for vertically integrated μLED systems. **4. Expected use** The technologies developed in this project will enable a new class of **LED-on-Silicon (LEDoS) platforms** that directly address the efficiency, speed, and integration challenges of conventional μLED architectures. **4.1. CMOS-integrated LEDoS via membrane transfer** The matured exfoliation and bonding processes will enable direct integration of μLED membranes with silicon CMOS circuitry through layer transfer. Unlike conventional pick-and-place approaches, this membrane-level integration allows seamless co-integration of dense μLED arrays with advanced electronic drivers, providing a practical pathway toward high-resolution, low-power microdisplays. **4.2. Enhanced efficiency and speed through backside accessibility** By rendering both sides of the μLED membrane accessible after transfer, the proposed platform enables fundamentally improved device architectures. Backside access facilitates vertical current injection and improved lateral current spreading, significantly reducing parasitic resistance and capacitance. As a result, the LEDoS platform is expected to exhibit higher efficiency and faster modulation speeds compared with conventional top–top-contact μLEDs, directly addressing RC-delay-related performance bottlenecks. **4.3. Resonant cavity integration for ultrafast operation** The accessibility of both the top and bottom surfaces of the μLED membrane further enables the integration of optical resonant structures, such as Fabry-Perot cavities, distributed Bragg reflectors (DBRs) and reflective mirrors. These resonant cavities can enhance light extraction, spectral control, and spontaneous emission rates, providing a route toward ultrafast μLED operation. Such capabilities are critical not only for advanced AR/VR display systems but also for emerging applications requiring high-speed optical modulation.
R&D Target Specifications	**Target specifications of UIUC's μLED membranes** • Optical Properties - Peak Wavelength: 460 nm ± 5 nm (Wavelength optimized for holographic waveguide input efficiency) - Spectral Bandwidth (FWHM): < 20 nm (Implementation of narrow-band emission via vertical cavity effect (Significantly reduced compared to conventional LEDs)) • Structural Properties - Total Epi Thickness: 500 - 1000 nm (Precise thickness control for resonance mode formation (p-GaN ~175 nm, MQW, n-GaN ~390 nm)) - Surface Roughness (RMS): < 1 nm (Ensuring compatibility for metal bonding processes) • Process Properties - Lift-off Yield: > 95 % (Crack-free transfer rate during the 2DLT (2D Layer Transfer) process) - Wafer Size: 2–4 inch (Ensuring large-area uniformity)
R&D Plan	**UIUC PI's R&D Plan** The goal of this R&D plan is to establish a scalable LED-on-Silicon (LEDoS) platform enabled by 2D materials–assisted epitaxy, addressing the key materials, device, and integration challenges. The work plan is organized into three tightly coupled tasks: (1) development of ideal 2D templates, (2) realization of ultrathin, high-efficiency μLED epilayers, and (3) exfoliation and heterogeneous integration with electronic backplanes. Together, these tasks form a coherent pathway from materials growth to system-level

demonstration.

Task 1. Development of 2D templates for 2D materials-assisted epitaxy

The objective of Task 1 is to establish **robust and scalable 2D templates** that enable high-quality III-N and III-V epitaxy while supporting subsequent exfoliation and transfer. We will investigate *c*-plane sapphire substrates and GaN(0001) templates as primary candidates for 2D-assisted growth, as each presents distinct advantages and trade-offs. Sapphire substrates, owing to their high ionicity, are favorable for remote epitaxial interactions, yet their large lattice mismatch with GaN poses risks to crystalline quality. In contrast, GaN templates (e.g., GaN-on-sapphire or GaN-on-silicon) enable straightforward remote homoepitaxy but typically suffer from high threading dislocation densities (TDDs) on the order of 10^8–10^{10} cm^{-2}.

We hypothesize that **2D interlayers can simultaneously act as dislocation filters and strain-relaxation mediators**, suppressing the propagation of threading dislocations while enabling interfacial sliding–assisted strain relaxation. Building on the PI's recent demonstration of wafer-scale BN growth on GaN substrates, we will systematically investigate BN growth on both sapphire and GaN templates. Key variables will include BN thickness, crystallinity, coverage uniformity, and growth temperature, and their impact on subsequent III-N or III-V nucleation, strain evolution, and defect formation.

As a risk mitigation strategy, if BN proves suboptimal under certain conditions, alternative 2D templates, including amorphous carbon (a-C), graphene, and transition metal dichalcogenides (TMDs), will be explored. These materials have previously shown promise in enabling remote or quasi–van der Waals epitaxy and provide additional flexibility in tailoring interfacial energetics.

Task 2. 2D materials-assisted epitaxy of ultrathin, high-efficiency μLED

The objective of Task 2 is to realize **ultrathin (<1 um total thickness), high-efficiency μLED epilayers** across the visible spectrum using the optimized 2D templates developed in Task 1. Building on the PI's prior demonstrations of blue and green InGaN MQW μLEDs grown on BN templates, this task will focus on further enhancing internal quantum efficiency (IQE), extending emission into the red spectral range, and minimizing buffer-layer thickness.

Using the prepared 2D templates, III-N and III-V nucleation and buffer layers will be grown by MOCVD, exploring both *in-situ* approaches (growth of BN, buffer, and LED layers within a single reactor run) and *ex-situ* approaches (separate growth steps). Particular emphasis will be placed on exploiting the **extremely low diffusion barriers on 2D surfaces**, which are expected to enable high-quality yet ultrathin buffers. With the nucleation conditions, we will fine-tune the growth of MQW structures. The strain relaxation enabled by 2D templates provides a unique opportunity to access high-indium-content InGaN without dislocation formation. If long-wavelength (e.g., green or red) LED membranes are necessary, advanced pulsed growth techniques—such as flow-modulated epitaxy and metal-modulated epitaxy—could be employed. These approaches enable lower effective growth temperatures while maintaining compositional uniformity. The combined use of 2D-enabled strain relaxation and advanced growth modulation is expected to yield high-efficiency MQWs for μLEDs.

Task 3. Exfoliation and heterogeneous integration with CMOS backplanes

The objective of Task 3 is to develop **scalable exfoliation and integration processes** that transform the ultrathin μLED epilayers into functional LEDoS platforms. Leveraging the weak van der Waals bonding at the 2D interface, we will establish deterministic, wafer-scale exfoliation of μLED membranes with minimal mechanical or thermal stress, in stark contrast to conventional lift-off or pick-and-place techniques.

Following exfoliation, μLED membranes will be bonded to silicon CMOS or TFT backplanes using bonding strategies compatible with ultrathin membranes, such as hybrid bonding or metal-assisted bonding. This membrane-level transfer enables direct integration of dense μLED arrays with driving circuitry while providing **full backside access to the epilayer**. Such access facilitates improved current spreading, reduced parasitic resistance and capacitance, and mitigation of RC-delay limitations that plague conventional top–top-contact μLEDs.

	In addition, access to both sides of the μLED membrane enables the integration of optical resonant structures, including AlGaN/GaN or dielectric-based DBRs. These resonant cavities will be explored as a route to enhance light extraction efficiency and enable ultrafast modulation. The performance of integrated μLEDs will be evaluated through measurements of EQE, modulation bandwidth, emission profiles, and device uniformity, providing quantitative feedback for iterative optimization.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The UIUC research team will act as the provider of the core source material (Epi-wafer) in this joint project, responsible for manufacturing and quality assurance of detachable ultrathin LED epi-wafers. Rather than providing off-the-shelf wafers, we will perform the role of designing and growing key materials tailored to determine the yield of the partner's 3D integration process and the optical performance of the final display. Utilizing our 2D-assisted epitaxy technology, we will supply high-quality ultrathin wafers in a "Ready-to-Transfer" state, allowing the partner to bond them immediately to circuit boards without complex substrate removal processes and will continuously provide material property data and technical guidance. - Application of vdW Epitaxy Source Technology: Provision of detachable epi-wafer manufacturing technology based on h-BN buffer layers. - Customized Epi-Wafer Design & Manufacturing: Precision epi-growth considering 460 nm resonant wavelength control and compatibility with the partner's process (Bonding). - Material Quality Assurance & Analysis: Evaluation of crystallinity/optical properties and data provision using specialized equipment (XRD, TEM, PL).
Expected R&R of Korean SME Partner	The Korean partner is expected to realize a **"CMOS-driven Monolithic 3D Integrated Holographic Waveguide Compatible Light Engine"** utilizing the ultrathin epi-wafers supplied by UIUC. To ensure the successful realization of ultra-high-resolution holographic displays, ideal partner would possess specific infrastructure and technical capabilities meeting the following quantitative criteria. The partner will develop the full process cycle from custom CMOS design to wafer-level optic integration. An example of the expected R&R is below, although it is not limited to the specified R&R. • Custom Active-Matrix CMOS Backplane Design & Fab: - Design and fabrication of a custom Si CMOS backplane (utilizing 130 nm or advanced node) optimized for μLED driving. - Develop a active-matrix pixel circuit capable of 16-bit scale control via Hybrid PAM/PWM driving, ensuring linear intensity modulation at low currents (< 1 nA). - The backplane process must guarantee a W-via surface planarity (CMP) of < 2 nm RMS to ensure void-free bonding interfaces. • Wafer-Level Monolithic 3D Integration: - Execution of high-yield eutectic bonding processes at fine pitches (< 7 μm pixel pitch). - Demonstrate lithographic capabilities with an overlay accuracy of < 600 nm to align the epi-membrane with CMOS nodes over a full wafer scale (2-4 inch), eliminating the alignment errors (> 2 μm) typical of conventional chip transfer. • Post-Bonding Device Fabrication & Cavity Control: - Execution of high-aspect-ratio dry etching for mesa isolation (< 1 μm spacing) and conformal sidewall passivation (e.g., ALD Al_2O_3) to suppress leakage current below 10 pA/pixel. - Control the optical cavity length during electrode formation to maintain a spectral width (FWHM) of < 20 nm at the center wavelength of 460 nm. • Advanced Optic Simulation & Beam Shaping Integration: - Design and integration of Microlens Arrays (MLA) or Meta-lenses with a high fill factor (> 90%).

	- Achieve a collimated beam divergence of < 10° (Half-angle) through precise alignment of the lens focal point with the μLED aperture (alignment tolerance < 200 nm). - Comprehensive simulation capabilities to construct and analyze optical systems, ranging from macro-optics (e.g., Laser, LED ray tracing) to nano-photonics using electromagnetic wave simulations (e.g., FDTD). • Holographic Module Design and Evaluation: - Design and fabricate holographic waveguide lens - System-level integration with wavelength-selective holographic waveguides, demonstrating compatibility with various guiding channel materials (e.g., Polymer, High-index Glass). - Demonstrate that the image transmitted through the waveguide retains sufficient brightness and sharpness to be clearly visible to the naked eye, ensuring high-fidelity visual perception for AR applications.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	• Technological Achievements - Vertical-Cavity Ultrathin μLEDs: We aim to realize ultrathin (~600 nm) μLED membranes featuring a vertical Fabry-Perot cavity. This architecture, achievable only through our unique van der Waals epitaxy, inherently narrows the spectral linewidth (< 20 nm). This provides a critical optical advantage for holographic waveguide coupling efficiency, overcoming the fundamental spectral limitations of conventional thick-film LEDs. - Revolutionary "Lithographic-Precision" Integration: By enabling the direct, damage-free transfer of continuous epi-membranes onto CMOS backplanes, our approach replaces stochastic "pick-and-place" assembly with wafer-level lithographic alignment. This process innovation eliminates alignment errors entirely, enabling the fabrication of ultra-high density μdisplays (> 5,000 PPI) required for immersive AR experiences. • Industrial and Economic Impact - Securing Foundational IP for AR Light Engines: By securing core patents for "Ready-to-Transfer" epi-wafers that do not require complex laser lift-off (LLO) processes, we position ourselves as a dominant player in the next-generation AR supply chain. This establishes a high entry barrier against competitors relying on conventional low-yield integration methods. - Synergistic Ecosystem Creation: This project creates a powerful synergy by combining UIUC's unrivaled source material expertise with the Korean SME's unique technology for 3D vertical integration of CMOS on devices. This vertical integration will accelerate the commercial deployment of lightweight, high-performance AR glasses, capturing a significant share of the rapidly expanding metaverse market.

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6. References

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